St. Philomena's College (Autonomous), Mysore

PG Department of Studies and Research in Chemistry

Question Bank (Revised Curriculum 2020 onwards)

FirstYear – First Semester (2020 -21 Batch)

Course Title (Paper Title): Basic Physical Chemistry.

QP Code: 85124

Unit	SI. No	Questions	Marks
1	1	Give the limitations of the first law of thermodynamics. Mention the importance of second law.	02
1	2	Write the entropy conditions for reversible and irreversible processes.	02
1	3	Define the term fugacity. How is it related to chemical potential?	02
1	4	What is standard state? Define the standard states for a gas and a liquid.	02
1	5	Write any two Maxwell's relations.	02
1	6	The Helmholtz free energy is also termed as work function.Give reason.	02
1	7	What does the activity coefficient signify? Can it have value greater than unity?	02
1	8	The vapour pressure of an aqueous solution of 0.5M KNO_3 at 100°C is 750 Torr while the vapour pressure of pure water at the same temperature is 760 Torr. Calculate the activity of water in KNO_3 solution.	02
1	9	Define chemical potential and mention its significance.	02
1	10	Differentiate between molar volume and partial molar volume.	02
1	11	Define the term activity and activity co-efficient.	02
1	12	State IIIrd law of thermodynamics.	02
1	13	Explain the significance of chemical potential.	02
1	14	The activation energy of a reaction at a given temperature is found to be 2,303RT J/mol. The ratio of the rate constant to the Arrhenius factor.	02
1	15	Deduce Gibb's-Helmholtz equation.	02
1	16	Write the variation of activitywith temperature.	02
1	17	An electrolyte has an activity 0.02 and activity co-efficient 0.92. Calculate	02

		the concentration.	
1	18	How does fugacity varies with pressure.	02
1	19	How does fugacity varies with temperature.	02
1	20	Write theexpressionfor fugacity and give its physical significance.	02
1	21	Which of the following conditions in a reaction is spontaneous? Explain. 1. ΔH= +ve., ΔS= +ve 2. ΔH=+ve., ΔS= -ve	02
1	22	If the concentration of the solution is 0.01M and its activity co-efficient is 0.92, what is the value of activityof that solution.	02
1	23	Provide the combined statement of I and II law of thermodynamics.	02
1	24	What is chemical potential? Indicate its significance.	02
1	25	Differentiate between Gibb's and Helmholtz free energies.	02
1	26	For the electrolyte $BaCl_2$, how is the mean ionic activity co-efficient related to the single ion activity co-efficient of the Ba^{2+} and Cl^- ions?	02
1	27	All spontaneous processes tend to increase the entropy of the universe. Explain.	02
1	28	The vapor pressure of an aqueous solution of 0.2M KCl at 100C is 749.7 Torr while the vapor pressure of pure water at the same temperature is 760 Torr. Calculate the activity of water in KCl solution.	02
1	29	Differentiate between fugacity and activity.	02
1	32	Chemical potential is a state function and an intensive property. Justify	02
2	30	How does primary salt effect differ from secondary salt effect?	02
2	31	Show that 0.3M NaCl and 0.1M K_2SO_4 have identical ionic strength.	02
2	33	Outline the basic principles of pulse radiolysis in the study of fast reactions.	02
2	34	In the determination of order of a reaction, it is always advisable to investigate it in the initial stages of the reaction. Explain	02
2	35	What are reversible and irreversible electrodes?	02
2	36	What are the advantages of stopped flow technique over continuous flow	02

		technique?	
2	37	Indicate the superiority of transition state theory of reaction rates over collision theory of reaction rates.	02
2	38	The rate constants of a reaction at 550°C and 625°C are 1.0L/mol/s and 6.4 mol/s. Calculate E_a for the reaction.	02
2	39	Collision theory fails to explain chain reactions. Give reason	02
2	40	Give the principle of pulse radiolysis.	02
2	41	Write the expressions for the rate constant of a first order and second order reaction and explain the terms involved in it.	02
2	42	Outline the principle of isolation method of determination of order of a reaction.	02
2	43	In a zero order reaction, the initial concentration of the reactant is 0.5N. If the rate constant is 8.05×10^{-4}mol/lit/s. Calculate the half-life of the reaction.	02
2	44	The half-life of a reaction is halved as the initial concentration of the reactant is doubled. Determine the order of the reaction.	02
2	45	The reaction H_2+ Br_2---- 2HBr is considered as a chain reaction. Justify	02
2	46	Define consecutive reactions with examples.	02
2	47	Reaction between glucose and water doesn't show any ionic effect on the rateof reaction. Why?	02
2	48	For a zero order reaction, the initial concentration of the reactant is 0.4N. If the rate constant is 1.05×10^{-3}mol/lt/s. Calculate the half-life of the reaction.	02
2	49	Define parallel reactions with an example.	02
2	50	Define relaxation effect.	02
2	51	Define contact time and write its equation for fast reactions.	02
2	52	Depict the behavior of Langmuir's equation at the limits of low and high pressure.	02
2	53	For a zero order reaction, the concentration of the reactant at 40min is 0.1m. Calculate the rate constant.	02
2	54	What is consecutive reaction? Give an example.	02

2	55	Write the equation for contact time and relaxation time.	02
2	56	Describe salt effects.	02
2	57	Explain cage effect with an example.	02
2	58	Outline the principles of pulse radiolysis in the study of fast reactions.	02
2	59	Calculate the contact time for the reaction if the rate of reaction is 2×10^{4} mole/dm^3/s and the extent of reaction is 0.001 mol/dm^3	02
2	60	What are the rate equations for the first and zero order reaction?	02
2	61	For the decomposition of acetone dicarboxylicacid, the rate constant is 2.46×10^{-5}/s at 273K and 1.61×10^{-3}/s. Calculate the E_a of the reaction.	02
2	62	What is meant by consecutive reactions? In what way it is different from the opposing reaction.	02
2	63	Write the postulates of Arrhenius theory of electrolytes.	02
2	64	What is polarogram? Indicate its importance.	02
2	65	Differentiate between activation and concentration polarization.	02
2	66	What are the forces acting on the free mobility of ions? Write the related mathematical expressions.	02
2	67	Using Debye-Huckel extended equation, calculate the mean ionic activity co-efficient of 0.2M aqueous solution of barium chloride.	02
2	68	How symmetry effect retard the mobility of ions?	02
2	69	Explain abnormal transport number with an example.	02
2	70	Calculate the mean ionic activity co-efficient of 0.01M aqueous solution of $BaCl_2$ at $30^{o}C$.	02
3	71	Explain ionic atmosphere in solution.	02
3	72	How does LJP arises?	02
3	73	Explain the role of gelatin and passing hydrogen gas through solution in polographic experiment.	02
3	74	Define the term concentration polarization.	02
3	75	The ionic conductance of sodium and bromide ions of 0.1N solution of NaBr	02

		are 43/ohm/cm and 37/ohm/cm, respectively. Calculate the transference number of the cation and anion in this solution.	
3	76	How do you verify the valdity of Debye-Huckel limiting electrolytes?	02
3	77	What are the limitations of Arrhenius theory of electrolytes?	02
3	78	What are true and potential electrolytes? Give two examples.	02
3	79	Why hydrogen gas is passed in the poloragraphy experiment?	02
3	80	Distinguish between reversible and irreversible cell.	02
3	81	How ohmic over potential produces?	02
3	82	Calculate the ionic strength of the solution containing 0.1M K_2SO_4 and 0.01N NaCl.	02
3	83	If transport number of Na^+ is 0.258 at 20°C, What will happen if the temperature is increased?	02
3	84	What are true and apparent transport numbers?	02
3	85	Define concentration and activation over voltage.	02
3	86	What are reversible electrodes? Give example.	02
3	87	The mobilities of H^+ and Cl^- at 25°C in water are 3.625×10^{-7} $m^2/s/v$ and $7.91\times 10^{-8} m^2/s/v$, calculate the transport number of H^+.	02
3	88	Indicate the role of gelatin and passing hydrogen gas through the solution in polarographic experiment.	02
3	89	What are elecrtrolytic cells? Explain with an example.	02
3	90	Differentiate between a battery and a cell.	02
3	91	Write the Debye-Huckel-Onsager equation, explain the terms involved in the equation.	02
3	92	Why gelatin is added during poloragraphy experiment?	02
3	93	What is liquid junction potential? How to eliminate LJP.	02
3	94	Give the effect of P^H on overvoltage.	02
3	95	What is abnormal transport number?	02

3	96	What is ohmic overvoltage?	02
3	97	Define electrical double layer.	02
3	98	Differentiate between true and apparent transport numbers.	02
3	99	Define polarization and overvoltage.	02
3	100	Write the difference between true and abnormal transport numbers.	02
3	101	Write the Debye-Huckel limiting equation and what is the purpose of adding supporting electrolyte while measuring $E_{1/2}$.	02
3	102	Define electrolytic polarization	02
3	103	State Onsager reciprocality relation in the mathematical form.	02
3	104	What is ionic atmosphere? How is the radius of ionic atmosphere determined?	02
3	105	Explain polarization	02
3	106	Calculate the LJP at 25°C between two solutions of HCl with mean ionic activities 0.01 and 0.1(t_{H^+} =0.828)	02
3	107	What is electrophoresis effect?	02
3	108	Write any two difference between primary and secondary batteries.	02
3	109	How does liquid junction potential arises.	02
3	110	Explain ionic atmosphere in solution.	02
3	111	What is the effect of current density on overvoltage?	02
3	112	Define abnormal transport number with example.	02
1	113	Give the physical significance of entropy and discuss the entropy changes in an ideal gases.	05
1	114	The volume of the solution is given by the relation, V=1000.38 + 20.563m + 2.024m^2-0.24m^3. Calculate the partial molar volume of the solute and solvent of 0.4m solution.	05
1	115	Give the significance of fugacity? Discuss the theoretical basis for the	05

		determination of fugacity of real gases using solubility measurements.	
1	116	Deduce Gibbs-Helmholtz relation and how entropy varies with temperature.	05
1	117	Define fugacity. How is fugacity determined by graphical method?	05
1	118	Give the physical significance of partial molar free energy. Obtain the expression $\mathcal{E}_{\mu}idn=0$	05
1	119	Define Gibbs free energy. How does it vary with temperature and pressure? Explain.	05
1	120	Define Partial molar volume. Outline the principle of intercept method of determination of partial molar volume.	05
1	121	State Nernst heat theorem and derive any two Maxwell's relations.	05
1	122	Define Chemical potential. Derive the expression for variation of chemical potential with temperature.	05
1	123	At 25°C, the density of a 50% by mass of ethanol water solution is 0.914g/cm^3. Given that the partial molar volume of water in the solution is 17.4 cm^3/mol. Calculate the partial molar volume of ethanol.	05
1	124	Outline the theoretical basis for the determination of activity co-efficient by measurement of depression in freezing point.	05
1	125	Discuss the determination of absolute entropy of a gas at 25°C and one atmospheric pressure from third law of thermodynamics.	05
1	126	Give an account of the thermodynamic criteria of equilibrium and spontaneity.	05
1	127	Show that entropy is a measure of unavailable energy.	05
1	128	Define molar and partial molar volume? Explain the density method for the determination of partial molar volume of solvent.	05
1	129	Discuss the application of third law of thermodynamics for the calculation of absolute entropies.	05
1	130	At 25°C, the density of a 50% by mass of ethanol-water solution is 0.914g/cm^3. Given that the partial molar volume of water in the solution is 17.4cm^2/mol. Calculate the partial molar volume of ethanol.	05
1	131	How do you determine the activity co-efficient by EMF method?	05

1	132	Describe the experimental method for the determination of the fugacity of a gas.	05
1	133	How does free energy vary with temperature and pressure?	05
1	134	What is partial molar volume? Discuss the determination of it by using density measurements.	05
1	135	Derive Duhem-Margules equation.	05
1	136	State Nernst heat theorem. How does it lead to the third law of thermodynamics?	05
1	137	How can the absolute entropy of gases and liquids be determined by applying third law of thermodynamics.	05
1	138	Define fugacity. Support one method to determine fugacity of a gas.	05
1	139	Give an outline of solubility method used for determining the activity co-efficient of AgCl.	05
1	140	Calculate the mean ionic activity co-efficient of 0.005molar aqueous solution of $K_3Fe(CN)_6$ at 25°C (A= 0.509 for water at 25°C)	05
1	141	Outline the theoretical basis for the determination of activity co-efficient by emf measurements.	05
1	142	The volume of the solution is given by the relation, V(1000)= 1001.80+ 24.522m1/2 + 3.486m+ 40.452m3/2. Calculate the partial molar volume of the solute and solvent of 2 molar solution.	05
1	143	Explain the intercept method of determination of the partial molar volume of solvent and solute.	05
1	144	State Nernst heat theorem. Discuss its applications.	05
1	145	Give a brief account on the experimental determination of E_a in Arrhenius equation.	05
1	146	Outline the theoretical basis for the determination of activity coefficient by measurement of depression in freezing point.	05
1	147	What is meant by the term Gibbs free energy? How does it vary with temperature and pressure?	05
1	148	How is activity co-efficient is determined by vapor pressure method.	05

1	149	State and justify the criterion for equilibrium and spontaneity.	05
1	150	Calculate the equilibrium constant for the dissociation of water at 2300K and 1.0 bar pressure, if the gibb's free energy change is 118.0kJ/mol	05
1	151	Formulate Gibbs-Duhem equation. How do you evaluate the activity coefficient of a component using Gibbs-Duhem equation?	05
1	152	On the basis of third law of thermodynamics how do you determine absolute entropy of a gas?	05
1	153	If the volume of the solution is given by the relation, V= 1000.38 + 20.563m + 2.024m^2 - 0.24m^3. Calculate the partial molar volume of solute and water at 0.5molal concentration.	05
1	154	What is partial molar volume of solute? From intercept method how do you evaluate the partial molar volume of solute in a binary solution?	05
1	155	Formulate Duhem-Margules equation	05
1	156	Calculate change in Gibbs free energy when 2 moles of an ideal gas expands isothermally and reversibly at 300K from a pressure 10atm to 5atm.	05
1	157	State and explain Nernst heat theorem and its applications.	05
1	158	What is partial molar volume? How it is determined by density measurements.	05
1	159	Derive an expression for absolute entropy from third law of thermodynamics.	05
1	160	Obtain the expression for the variation of fugacity with temperature and pressure.	05
1	161	What is meant by activity and activity co-efficient of a substance? The activity of 2.5mol of a substance changes from 0.05 to 0.35. What would be the change in its free energy?	05
1	162	Discuss the theoretical interpretation for the determination of fugacity by graphical method.	05
2	163	Explain secondary salt effects for kinetics of reaction in solution.	05
2	164	Give an account of activated complex theory and derive the expression for the rate constant.	05
2	165	Write a brief note on potential energy surfaces and cage effect.	05

2	166	Discuss the effect of pressure and volume on the rate of reaction.	05
2	167	Give the significance of activation energy and gibb's free energy in chemical kinetics.	05
2	168	For a first order reaction, the energy of activation is 10k/cal/mol. If the rate constant at 40°C is 4×10^{-4}/sec.Calculate the rate constant at 30°C	05
2	169	Write the assumptions of activated complex theory and its application in the calculation of activation energy.	05
2	170	Discuss the double sphere model for the reaction between ions in solution.	05
2	171	Outline the method of studying fast reactions by stopped flow technique.	05
2	172	Give a comparative account of collision theory and the activated complex theory.	05
2	173	Explain the effect of pressure on the rate of reaction.	05
2	174	Describe the influence of dielectric constant of the medium on the rates of reaction and classical thermodynamics.	05
2	175	A reaction has energy of activation of 14.6kcal/mol. If the rate constant of a first order reaction at 45°C is 0.292/min, calculate the half-life of the reaction at 60°C	05
2	176	Explain the mechanism of oscillatory reactions with a suitable examples.	05
2	177	Out line the shock tube method for the study of fast reactions.	05
2	178	What is relaxation time? Calculate the relaxation time for 0.1M acetic acid at 25°C. The K_f and K_r values are 7.8×10^{5}/sec and 4.5×10^{10} dm^3/mol/sec.	05
2	179	Derive the Bronsted-Bjerrum equation for the primary salt effect on the rate of reaction.	05
2	180	Give a detailed account of the collision theory of reaction rates along with its limitations.	05
2	181	A reaction is 20% complete in 15min at 40°C and in 3min at 60°C. Calculate the E_a and other thermodynamic parameters.	05
2	182	Deduce general expression for a branched chain reaction.	05
2	183	What are side reactions? Discuss the kinetics of side reactions.	05

2	184	The reaction has energy of activation of 65.79kJ/mol and if the rate constant of a first order reaction at 45°C is 0.211/min. Calculate the half-life of the reaction at 60°C	05
2	185	Discuss the collision theory of bimolecular reaction rates and explain the significance of frequency factor and steric factor.	05
2	186	Explain flash photolysis technique of studying fast reactions.	05
2	187	Discuss the general mechanism of branched chain reactions.	05
2	188	Give comparative account of collision theory and the activated complex theory.	05
2	189	Calculate the frequency factor for the reaction of H_2 with I_2 at 300°C. The rate constant for the reaction and energy of activation are 0.85dm^3/mol/sec and 38.5kcals/mol (R= 1.987cals/deg/mol).	05
2	190	Discuss the variation in the rate constant of an ionic reaction with ionic strength of the solutions.	05
2	191	Outline the principle of relaxation method of studying fast reactions.	05
2	192	Obtain the expression for rate constant of reversible reactions which are first order in both directions.	05
2	193	Discuss the double sphere model of reaction between ions in solution.	05
2	194	With neat diagrams, explain the continuous and stopped flow techniques of studying fast reactions.	05
2	195	Derive the expression for rate constant of a reaction based on the activated complex theory and discuss its applications to reactions in solution.	05
2	196	Calculate the energy of activation of a reaction whose rate constant at 27°C gets doubled for 10°C rise in temperature.	05
2	197	What is relaxation time? How kinetics of fast reactions are studied by temperature and pressure jump methods?	05
2	198	Write a note on primary salt effect and potential energy surface.	05
2	199	Give an account of Lindeman's theory of unimolecular reactions.	05
2	200	Discuss the kinetics of parallel reactions.	05
2	201	Give an account of continuous and stopped flow techniques in the study of	05

		fast reactions.	
2	202	Give the statistical formulation of activated complex theory of reaction rate.	05
2	203	Obtain a rate expression for the kinetics of consecutive reaction.	05
2	204	Explain briefly the theoretical calculations of activation energy.	05
2	205	Give an account of shock tube method for the study of fast reactions.	05
2	206	What are parallel reactions? Discuss the kinetics of parallel reactions.	05
2	207	Discuss in details the activated complex theory of reaction rates. In what way this theory is superior to collision theory	05
2	208	Explain the mechanism of bimolecular reactions on the basis of Lindeman's theory.	05
2	209	Discuss the double sphere model of effect of dielectric constant on the kinetics of reactions in solution.	05
2	210	Outline the relaxation technique in the study of fast reactions.	05
2	211	Discuss the theoretical calculation of energy of activation in a bimolecular reaction using potential energy surface.	05
2	212	Discuss briefly double sphere model for the reaction between two ions in solution.	05
2	213	Discuss Hinshelwood theory of unimolecular reactions.	05
2	214	How do you compare the collision theory of reaction rates with the transition state theory?	05
2	215	Obtain the rate expression for the reaction of the type A-------B-------C with the rate constants k_1 and k_2.	05
3	216	Calculate the thickness of ionic atmosphere in 0.1N solution of a univalent electrolyte in ethyl alcohol(D= 24.3) at lab temperature.	05
3	217	Write a brief note on cage effect and secondary salt effect.	05
3	218	Calculate the value of the liquid junction potential at 25°C between two solutions of HBr having mean ionic activities of 0.011 and 0.001, respectively. The t^+ of H^+ ion in HBr can be taken as 0.68.	05
3	219	How do you test the validity of Debye-Huckel Onsager conductance	05

		equation?	
3	220	Discuss the principles and applications of polarography.	05
3	221	Deduce the Butler-Volmer equation for irreversible electrode processes.	05
3	222	Define and differentiate with suitable examples between ohmic, concentration and activation over voltages.	05
3	223	State and explain the factors effecting Debye-Huckel Onsager conductivity equation.	05
3	224	State the Arrhenius and Debye Huckel theory of strong electrolytes.	05
3	225	What is liquid junction potential and explain a method of determining it.	05
3	226	Deduce Debye -Huckel limiting law	05
3	227	How electrophoretic and relaxation forces retard the free mobility of ions? Deduce the expression for Debye-Huckel-Onsager conductance equation.	05
3	228	The EMF of the concentration cell with transference. Pt, H_2(1atm)/HCl (a^+-=0.0175), H_2 (1atm)Pt is 0.028V at 25°C. The EMF of the corresponding cell without transference is 0.017V.Calculate the LJP and the transference number of H^+ ion.	05
3	229	Discuss the quantitative aspects of true and apparent transference numbers.	05
3	230	What are reversible and irreversible electrodes? Explain with suitable examples.	05
3	231	Derive Onsager's reciprocity relation	05
3	232	Derive the Debye-Huckel-Onsager conductance equation.	05
3	233	State and explain the Debye-Huckel equation for appreciable concentration.	05
3	234	How is polarographic technique used for qualitative and quantitative estimation of metal ions.	05
3	235	What are secondary batteries? With a suitable example, discuss the construction and working principle of a secondary battery.	05
3	236	With illustrative example outline the theoretical basis for the determination of transference number by the measurement of EMF.	05
3	237	What is meant by ionic strength of a solution? Calculate the ionic strength of a solution prepared by mixing 50ml of 0.2M KNO_3, 20ml of 0.15M K_2SO_4	05

		and 30ml of 0.005M $Cu(NO_3)_3$.	
3	238	What is Debye-Huckel reciprocal length? Deduce the expression for electrostatic potential of the ionic atmosphere.	05
3	239	Small changes in η(eta) produces large changes in i-prove this concept by Butler-Volmer equation.	05
3	240	Discuss the influence of temperature and concentration on transference number.	05
3	241	Explain briefly experimental determination of over voltage	05
3	242	Write the experimental determination of liquid junction potential.	05
3	243	Write a brief note on true and apparent transport number.	05
3	244	Deduce Butler-Volmer equation for irreversible electrode process.	05
3	245	Explain half-wave potential and its application in qualitative and quantitative analysis.	05
3	246	What is concentration overvoltage? Explain the experimental determination of over voltage.	05
3	247	Formulate Debye-Huckel-onsager conductance equation and explain the experimental verifications.	05
3	248	How do you determine transference number by Hittorf's method?	05
3	249	Explain the effect of temperature and PH on over voltage.	05
3	250	Derive Butler-Volmer equation.	05
3	251	How do you estimate metal ions qualitatively and quantitatively by polarographic method?	05
3	252	Explain reversible and irreversible electrodes.	05
3	253	Write a note on hydrogen-oxygen over voltage.	05
3	254	What is ionic atmosphere? Obtain the expression for electrostatic potential of the ionic atmosphere.	05
3	255	Calculate the value of the liquid junction potential at 25°C between two solutions of HCl having mean ionic activities of 0.01 and 0.001, respectively. The t^+ of H^+ ion in HCl can be taken as 0.78.	05

3	256	Deduce Debye-Huckel Onsager conductance equation. How do you test the validity of this equation?	05
3	257	Draw Tafel plot. Explain the significance of slope and intercept of Tafel plot.	05
3	258	Explain the influence of P^H and temperature on over voltage.	05
3	259	Explain the principle and applications of polarography.	05
3	260	Write a note on abnormal transport number.	05
3	261	What is liquid junction potential? Explain the experiment to determine the LJP.	05
3	262	Derive Debye-Huckel limiting equation for activity coefficients.	05
3	263	What is Half-wave potential? Explain qualitative and quantitative application of polarography.	05
3	264	Explain the experiment to determine the transport number of cation.	05
3	265	Write a note on Tafel equation and Effect of current density on over voltage.	05
3	266	Discuss Butler-Volmer equation and its significance.	05
3	267	How do you determine the transport number by EMF Method.	05
3	268	Derive Butler-Volmer equation.	05
3	269	Explain Hittorf's method for the determination of transport number.	05
3	270	Calculate λ_o for KCl solution.Given that the ionic molar conductance at infinite dilution are 7.352×10^{-3} Sm^2/mol for K+ and 7.634×10^{-3} Sm^2/mol for Cl-.	05
3	271	Explain the use of polarographic technique in the analysis of organic compounds.	05
3	272	Deduce an equation for diffusion current and stationary current and give a comparative account of these two equations.	05
3	273	Write the limitations of Arrhenius theory of strong electrolytes.	05
3	274	Differentiate between reversible and irreversible electrodes.	05
3	275	Calculate the radius of ionic atmosphere of a uni-univalent electrolytic solution of 0.1N at 20°C for nitrobenzene with d= 34.8 and acetone with d =	05

		20.7	
3	276	Explain Relaxation and electrophoretic effects.	05
3	277	Deduce Debye-Huckel limiting law	05
3	278	With illustrative example differentiate between true and apparent transference number.	05
3	279	Define and differentiate with suitable examples between ohmic,concentration and activation over voltages.	05
3	280	List various types of reversible and irreversible electrodes. Give examples each.	05
3	281	With illustrative example outline the theoretical basis for the determination of transference number by EMF Measurements.	05
3	282	Small changes in η(eta) produces large changes in 'i'- prove this concept by Butler-volmer equation.	05
3	283	Write a brief note on experimental determination of overvoltage.	05
3	284	The viscosity of water at 25°C is 0.00895 poise, and dielectric constant is 78.5. Calculate Onsager constant.	05
3	285	Calculate the mean ionic activity co-efficient of $BaSO_4$ electrolyte at a molarity of 0.002M in aqueous solution at 25°C	05
3	286	Calculate the mean ionic activity co-efficient of 0.001M aqueous solution of $K_3[Fe(CN)_6]$ at 25°C.	05
3	287	A 0.0001M solution of ferric sulphate $[Fe_2(SO_4)]$ was prepared using 0.0001M H_2SO_4 as solvent. Calculate the mean ionic activity co-efficient of two electrolytes at 25°C	05
3	288	Calculate the ionic strength of following solution 50ml of 0.2M KNO_3, 20ml of 0.1M K_2SO_4 and 30ml of 0.5M of $Cu(NO_3)_2$.	05
3	289	Calculate the ionic strength of the following solution 60ml of 0.1M KCl, 20ml of 0.2M K_2SO_4 and 20ml of water.	05
3	290	Calculate LJP at 23°C between the two solutions of HCl of mean ionic activities 0.1 and 0.01 respectively. The mean value of cationic transport number may be taken as 0.828.	05

3	291	The EMF of the concentration cell with transference. Pt, H_2(1atm)/HCl (a^+-=0.0091), H_2(1atm)Pt is 0.028V at 25°C. The EMF of the corresponding cell without transference is 0.0169V. Calculate the LJP and the transference number of H^+ ion.	05
3	292	Calculate the ionic conductance of H^+ion if the transport number is t_{H+}= 0.46 and equivalent conductance of HCl = 380/ohm/g/erg	05
3	293	If transport number of Na^+ is 0.394 and its ionic conductance is 35.7/ohm. Calculate ionic conductance of Cl-.	05
3	294	The ionic conductance of Na^+ and Br^- of 0.1N solution of NaBr are 43/ohm and 37/ohm respectively. Calculate the transport number of cation and anion inthis solution.	05
3	295	Calculate the thickness of ionic atmosphere in 0.05N solution of univalent electrolyte in the following solvents. a) Nitrobenzene =D=34.8 at 25°C b)Ethyl alcohol=D= 24.3 at 25°C.	05
2	296	The rate constant for a first order reaction is 4× 10-4/sec at 20°C and 12× 10-4/sec at 30°C respectively.Calculate the energy of activation.	05
2	297	The half life of a first order reaction is 20min at 20C and 15min at 30C respectively. Calculate energy of activation for the reaction.	05
2	298	50% of a first order reaction is complete in 23min.Calculate the time required to complete 90% of the reaction.	05
2	299	The equilibrium constant of a fast reaction is 4× 104, if the relaxation time is 2× 10^{-8} sec. Calculate forward and backward rate constants.	05
2	300	When [N_2O_5] =0.22M the rate of decomposition of N_2O_5 is 1.3× 10^{-4}mol/lit/sec. What is the value of K for first order reaction.	05
2	301	If the energy of activation of a reaction is 16kcal/mol when the temperature is raised from 25C to 35C, Calculate the specific reaction rate at higher temperature if its value at lower temperature is 2.5× 10-3 /min.	05

Question Paper Pattern- Model Question Paper

Q.P. Code: MCH20A104

St. Philomena's College (Autonomous), Mysore			
I Semester –Course M.Sc Final Examination			
Subject : Chemistry			
Title: Basic Physical Chemistry (HC)			
Time: 3 hrs			**Max Marks: 70**
PART-A			
1		**Answer any ten of the following:**	**10x2=20**
	a	How do you verify the valdity of Debye-Huckel limiting electrolytes?	2
	b	Why gelatin is added during poloragraphy experiment?	2
	c	What are true and apparent transport numbers?	2
	d	Show that 0.3M NaCl and 0.1M K_2SO_4 have identical ionic strength.	2
	e	Which of the following conditions in a reaction is spontaneous? Explain. 1. ΔH= +ve., ΔS= +ve 2. ΔH=+ve., ΔS= -ve	2
	f	Write any two Maxwell's relations	2
	g	For the decomposition of acetone dicarboxylic acid, the rate constant is 2.46×10^{-5}/s at 273K and 1.61×10^{-3}/s. Calculate the E_a of the reaction.	2
	h	What are the advantages of stopped flow technique over continuous flow technique?	2
	i	Collision theory fails to explain chain reactions. Give reason	2
	j	Calculate the contact time for the reaction if the rate of reaction is 2×10^4 mole/dm^3/s and the extent of reaction is 0.001 mol/dm^3	2
	k	Write the postulates of Arrhenius theory of electrolytes.	2
	l	How symmetry effect retard the mobility of ions?	2
PART-B			
		Answer any Five of the following:	**5 x 10= 50**
2	a	The ionic conductance of Na^+ and Br^- of 0.1N solution of NaBr are 43/ohm and 37/ohm respectively. Calculate the transport number of cation and anion inthis solution.	05
	b	Derive the expression for rate constant of a reaction based on the activated	05

		complex theory and discuss its applications to reactions in solution.	
3	a	Give an account of Lindeman's theory of unimolecular reactions	05
	b	Formulate Gibbs-Duhem equation. How do you evaluate the activity coefficient of a component using Gibbs-Duhem equation?	05
4	a	Define Partial molar volume. Outline the principle of intercept method of determination of partial molar volume.	05
	b	Discuss the double sphere model for the reaction between ions in solution.	05
5	a	What is relaxation time? How kinetics of fast reactions are studied by temperature and pressure jump methods?	05
	b	Obtain the expression for the variation of fugacity with temperature and pressure.	05
6	a	Deduce the Butler-Volmer equation for irreversible electrode processes.	05
	b	Discuss the principles and applications of polarography.	05
7	a	50% of a first order reaction is complete in 23min.Calculate the time required to complete 90% of the reaction.	05
	b	Derive the Bronsted-Bjerrum equation for the primary salt effect on the rate of reaction.	05
8	a	The volume of the solution is given by the relation, V(1000)= 1001.80+ 24.522m1/2 + 3.486m+ 40.452m3/2. Calculate the partial molar volume of the solute and solvent of 2 molar solution.	05
	b	State the Arrhenius and Debye Huckel theory of strong electrolytes.	05